

Longitudinal media review for audition-oriented violin practice

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High-level music training produces dense evidence: daily practice videos, repeated repertoire, changing technical constraints, and periodic performance goals. Most of that evidence is difficult to use because it is stored as unstructured media rather than as a longitudinal record. We report Curtis Media Review, a live AO Labs system that indexes public practice media, identifies practice candidates, records model-assisted section reviews, and converts the current state into a compact practice record. The checked live state on 7 May 2026 used the YouTube channel @nalalan as the primary source, indexed 207 public videos, marked 102 as practice candidates, marked 82 as long-form candidates, stored 20 media samples, retained 51 reviewed sections, and recorded 80 Curtis-focused findings. The first screen reports the repertoire-identification state, the score state, one immediate tip, and a source-linked highlight clip so the label can be checked against the actual playing. A growing repertoire list stores day-level progress entries, while a dated practice log keeps clips, detection status, and sample counts visible without exposing the full raw record by default. The newest layer is a practice-study packet: each confirmed practice day can carry a machine transcription, a public-domain score page, boxed target passages, a timed recording link, and passage-specific feedback. Possible labels, user-rejected false labels, weak evidence, and readiness scores inferred only from piece identification must remain distinguishable from confirmed repertoire and judged playing quality. The system uses GPT-5.5 for vision and text review and separate audio models for identification and verification, while preserving explicit limits: Curtis admission cannot be predicted from current samples, weak audio/video evidence is marked unjudged, and the system reports practice signals rather than admissions probability. The contribution is an experimental method for turning accumulated practice media into auditable feedback loops, progress history, and source-bounded coaching suggestions.

1 Introduction

Elite music preparation depends on repeated, deliberate practice and feedback over long time horizons (1, 2). The useful data are not only polished performances. They include ordinary practice sessions, failed takes, camera angle limitations, recurring technical problems, and changes in focus across days. A violinist recording daily practice can therefore produce a large longitudinal dataset, but the raw dataset does not automatically become an operating record.

Curtis Media Review addresses this problem as a live software system rather than as a portfolio page (3). The system indexes public practice media, classifies candidate practice logs, records technical observations, and keeps the current practice constraint visible. It is designed for the Curtis Institute goal, but it does not claim to estimate an acceptance probability. The current role of the system is narrower: observe practice evidence, retain history, separate judged from unjudged evidence, and return a small number of practice constraints that can be tested in subsequent recordings.

The system is also part of AO Labs' broader progress architecture. In May 2026, AO Progress was introduced as a shared ledger for cross-application state and long-term changes across public apps, papers, CV artifacts, Curtis practice state, Imagineer state, and Relay state (4). Curtis supplies one stream in that ledger: timestamped media and review evidence for musical development.

2 System state

The first production state used the live backend at <https://curtis.aolabs.io/api/curtis/media-status>. The checked state on 7 May 2026 reported the values in Table 1. These are operating facts, not outcome claims.

2.1 Media indexing

The source inventory stores platform, title, publication time, duration, view count, candidate reasons, media kind, and blocker state. Dated long-form practice logs are treated as high-value longitudinal records because the title contains a practice date and the duration preserves session-level evidence. Other videos are retained but not forced into the practice lane.

Table 1. Curtis production state on 7 May 2026.

Signal	Current value
Primary source	YouTube channel @nalalan
Inventory	207 indexed videos
Practice candidates	102 videos
Long-form candidates	82 videos
Media samples	20 captured samples
Reviewed sections	51 sections
Curtis-focused findings	80 findings
Latest dated practice log	5-3-26
Review model	GPT-5.5 for vision/text, separate audio models for audio evidence and piece verification
Confirmed labels	5/1 Haydn Symphony No. 94 IV; 5/2 and 5/3 Wieniawski Scherzo-Tarantelle
Training anchors	3 source-confirmed practice sources
Score-reference targets	3 configured
Practice-study packets	3 confirmed days with score snippets and timed practice links
Pitch/rhythm transcription	Configured; per-sample note events and fingerprints stored when media samples exist
Score-aligned windows	0; pitch/rhythm-to-score matching is not yet configured
Curtis-level completion	Withheld when the available evidence only identifies repertoire
Current focus	Audition-style setup consistency
Current constraint	Open strings only, mirror-check contact point
Boundary	Curtis admission cannot be predicted from current samples

2.2 Review evidence

The review layer stores section-level findings with a dimension, evidence source, judgment, and practice constraint. The current dimensions include tone, intonation, shifts, time, articulation, and audition delivery. Findings can be strong, needing work, or unjudged. The unjudged state is part of the system design: if the sound is not available, the hand is obscured, the camera angle hides a transition, or still frames do not support a timing claim, the system should not convert weak evidence into a confident critique.

3 First readout

The first live readout is experimental but usable. The strongest immediate value is that the system has already converted a large, timestamped practice archive into a queryable corpus with captured media samples and stored section reviews. The home screen now reduces the current state to a repertoire-identification state, a score state, one immediate tip, and the highlight clip that produced the label. This source-linked layout is deliberate: a detected title is not treated as self-evident unless the adjacent clip supports it, and a readiness percentage is not shown when the evidence only identifies repertoire. A visible repertoire list stores piece-level and day-level progress, and a dated practice log shows which video window was sampled for each practice day. A separate study packet now gives each confirmed day a low-density score/transcription view: one boxed score page, one clip link, one transcription state, and one passage-level feedback statement. Generic etude/caprice evidence remains outside the piece-title field, and possible or uncorroborated repertoire labels remain checkable against the source window. These statements are not generic pedagogy; they are extracted from the stored review findings, piece record, sample index, source corrections, public-domain score targets, and current progress plan.

4 Discussion

Curtis Media Review is intentionally conservative. It does not rank the player against applicants, infer admissions probability, or convert every video into a claim. Instead, it keeps a running technical record. This makes the system useful even before it becomes musically sophisticated: it can preserve practice volume, surface recurring constraints, identify missing evidence, and make subsequent recordings

Table 2. Current evidence classes and limits.

Class	Stored signal	Limit
Audio	Tone, intonation, shifts, time, articulation	Some audio reviews fail or cannot inspect the excerpt cleanly
Video	Setup, posture, bow lane, visible hand/bow evidence	Still frames can hide motion, sound, and transition accuracy
Metadata	Dates, duration, channel, view count, practice-candidate labels	Metadata cannot judge playing quality by itself
Progress plan	One focus, practice constraint, short session plan	The plan is a next experiment, not an outcome prediction

easier to compare against earlier ones. 79

The main open problem is repertoire identification and evidence quality. YouTube metadata can 80
identify candidate videos, and the current owner-sync path can provide direct audio/video samples, 81
but unclear camera framing, missing score/title context, and short excerpts still prevent reliable piece 82
naming. The backend therefore keeps generic labels out of the piece-title field, stores them as candidate 83
evidence, requires source-window evidence for confirmed titles, and marks weak evidence explicitly. 84
The current training ledger separates source-confirmed labels from independent audio-title matches, 85
machine pitch/rhythm windows, and score-aligned windows. It also stores reference targets for the 86
three confirmed source labels across two works so later passes can ask for section-level matching 87
instead of repeating broad title guesses. The transcription path converts sampled media to mono audio, 88
estimates monophonic violin pitch with a pYIN pitch track, segments stable note events, estimates 89
tempo, stores compact note-duration notation, and builds pitch-class, interval, rhythm, and contour 90
fingerprints for future matching. A stronger system should align those events against a symbolic or 91
optical score representation, identify movement sections or measure ranges, compare against reference 92
recordings, and then judge articulation, timing, tone, and tempo against the aligned passage. 93

The broader experiment is whether a practice system can make musical development legible over time. 94
A useful Curtis system should show whether tone, intonation, rhythm, articulation, shifts, memory, 95
endurance, and audition delivery are improving across dated recordings. It should also preserve the 96
source evidence behind each claim so that the dashboard does not become motivational text detached 97
from the actual playing. 98

5 Materials and methods

Backend

The backend is a FastAPI service deployed on Railway. It exposes `https://curtis.aolabs.io/api/curtis/media-status` for compact state, `https://curtis.aolabs.io/api/curtis/ops-check` for fuller operational state, sample indexes for duplicate-window avoidance, and run endpoints for scans, media probing, analysis, and coaching. Persistent runtime state stores sources, inventory, review findings, media samples, analysis runs, and scan history.

Source scan

The default public source is `https://www.youtube.com/@nalalan`. The scanner queries public video metadata, classifies videos by title, duration, and candidate reasons, and stores inventory records. Dated practice logs and long-form videos are treated as practice candidates. The scan does not itself inspect performance quality.

Model review

The model layer reviews sampled media sections where usable excerpts exist. GPT-5.5 is used for text and visual review, and separate audio models are configured for audio review and piece-title verification. Findings are sanitized so weak evidence terms such as obscured, not audible, or no clear evidence produce an unjudged state rather than an unsupported critique. Piece labels are also sanitized: coach reviews can store candidate evidence but cannot create confirmed repertoire titles, and the piece-identification layer must provide source-window evidence plus independent verification before it updates the current piece list. Same-video consensus can compare multiple sampled windows before accepting a title; for long date-stamped practice captures with later samples, early setup windows are excluded from confirmation, and confirmed active titles are versioned so older labels are rechecked when source rules change. User-rejected false labels are excluded from later passes only for the source where the correction applies, so later videos can still identify that repertoire when present. A source correction also invalidates older confirmed labels for that same source instead of promoting the next stale candidate, and repeated corrected false labels keep that source out of the confirmed repertoire list until a stronger acceptance path exists. Rejected proposed titles, top candidates, and verification titles are scrubbed from current API output, not only from confirmed piece records. The 5/1 forensic pass

rejected prior Paganini, Bruch, Mendelssohn, Brahms, Sibelius, Tchaikovsky, Wieniawski, Saint-Saens, Ravel, Bazzini, Ernst, Carmen, Zigeunerweisen, Bach Partitas, Kreisler Praeludium and Allegro, Sarasate family labels, Ysaye Sonata No. 3 Ballade, Mozart K. 216, Glinka Ruslan and Lyudmila, Strauss Till Eulenspiegel, Strauss Don Juan, Smetana Bartered Bride, Prokofiev Classical Symphony, Dvorak Carnival Overture, Shostakovich Symphony No. 5, Ravel Bolero, the first broad orchestral candidate batch, and the later overture, dance, and pops-orchestral candidate batch for that source. Alan then supplied the ground-truth source label: Haydn Symphony No. 94, IV. Finale, Violin I part. The system now treats that label as human-confirmed for 5/1, stops injecting new seeded guesses for the same source, and keeps Curtis-level completion unscored until judged playing evidence supports a score. Alan also supplied the 5/2 source label, Wieniawski Scherzo-Tarantelle, Op. 16, and then confirmed that all violin footage on 5/3 was the same piece. The training state now records all three source labels as anchors and reports zero score-aligned windows until pitch/rhythm extraction has matched a source window to a score section. Transcribed samples can now produce learned pitch/rhythm fingerprints, and future samples can receive a nearest-source hint when their extracted notes and rhythms resemble a prior confirmed source. Possible labels and model-only verified titles are treated as unidentified on the dashboard and repertoire list unless a source label is confirmed, so future failures appear as missing evidence rather than as wrong repertoire. Piece-identification output does not create a Curtis-level readiness percentage by itself; readiness scoring requires judged playing evidence. Dated practice entries are keyed by the source-video date rather than the later processing time, so historical uploads cannot masquerade as today's practice state. When no same-day sample exists, the current panel can show the latest source-dated evidence without marking it active today. The owner-browser sync uses the sample index to prioritize uncaptured public YouTube windows rather than re-uploading already reviewed excerpts, and the piece-identification pass can test multiple active moments inside each captured sample before withholding the title. Piece records retain day-level progress entries and source-window fields so the home screen can show the score state, a main highlight clip, and the longer-running piece record together. The practice-study builder joins those records to public-domain score PDFs, renders cached score pages through Poppler, overlays boxed target passages, and exposes a timed practice link plus machine transcription summary for each confirmed practice day.

Progress ledger

AO Progress tracks Curtis as part of a larger multi-application state record. The first progress snapshot included Curtis home, Curtis media state, other AO Labs apps, Imagineer state, Imagineer paper, Relay

state, and public project surfaces. Curtis contributes dated practice inventory, current focus, review status, and future trend signals.

6 Limitations

The current system cannot predict acceptance at Curtis. It also cannot replace a musician, teacher, jury, studio lesson, or audition process. It can be wrong when the source media are incomplete, when audio is unavailable, when video frames hide the physical action, or when model review fails. The useful claim is therefore limited: Curtis Media Review turns practice media into a persistent, source-bounded record that can support iteration over time.

References

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